

WEEK 5 – POWER BI HEALTHCARE DATA ANALYSIS

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INTRODUCTION:

Healthcare data contains information about patients, hospital admissions, medical conditions, treatments, insurance and billing. In this project, Power BI Desktop is used to prepare and organize the healthcare dataset.

Power Query is used for data cleaning, transformation and table creation. The data is divided into patient, admission and billing information to create a structured data model.

OBJECTIVES:

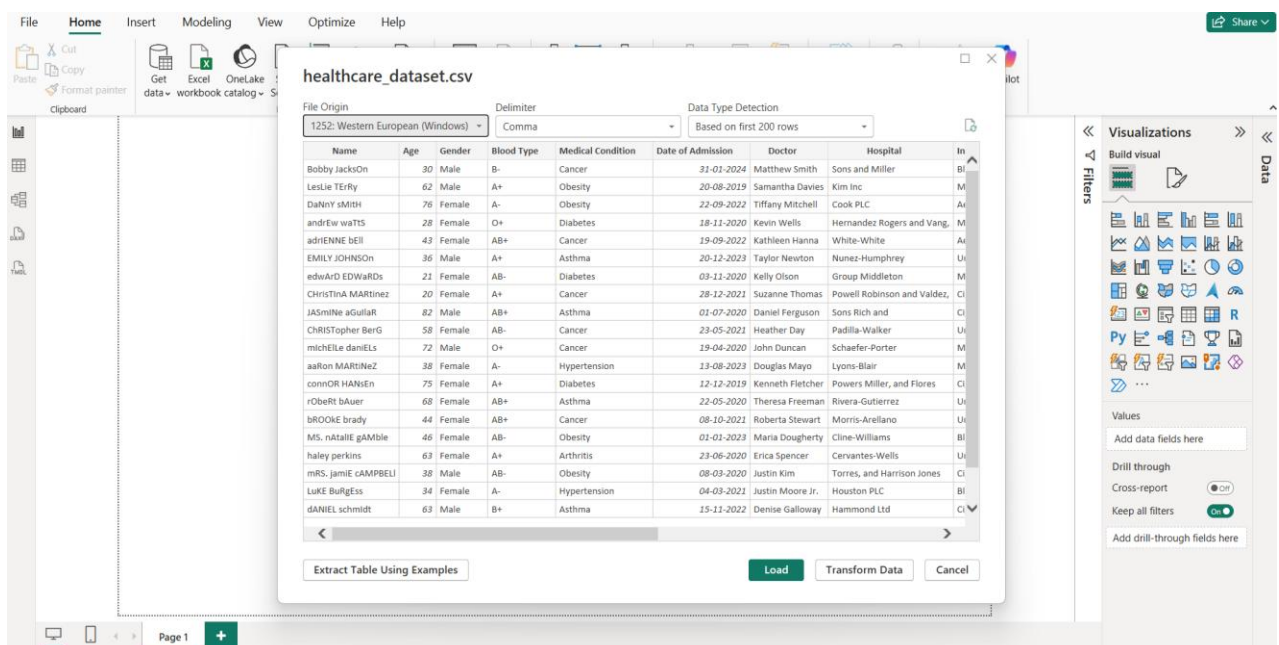
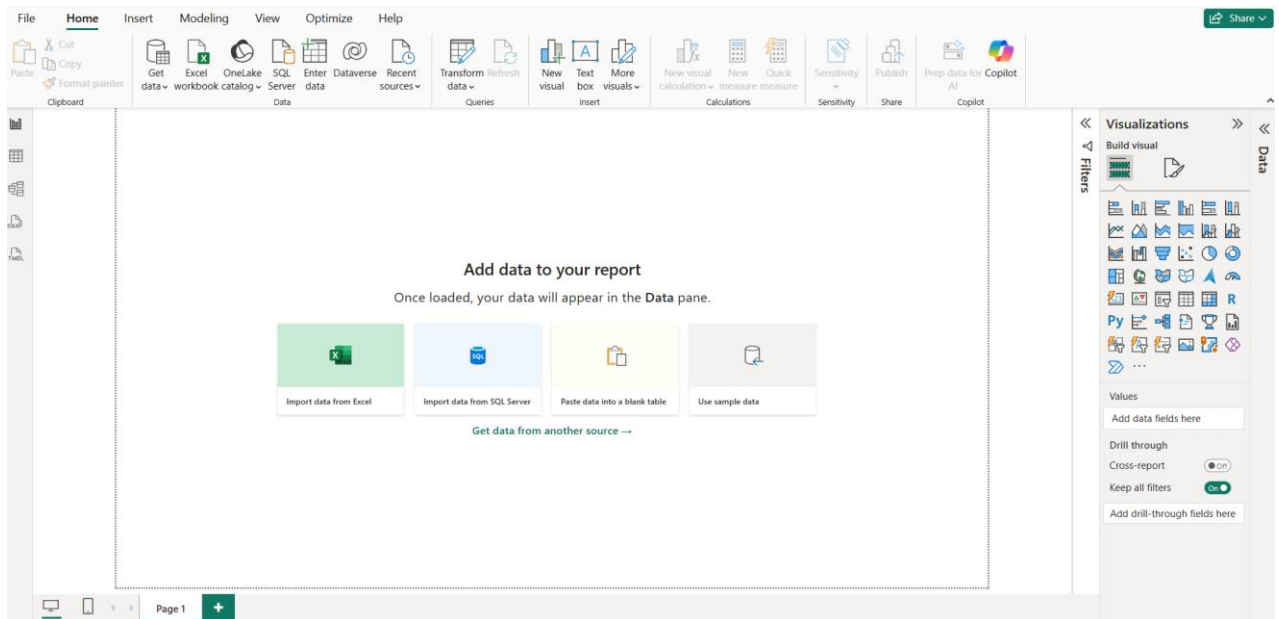
The main objectives of this project are:

- To import healthcare data into Power BI.
- To clean and standardize the dataset.
- To create meaningful age groups.
- To create a separate patient dimension table.
- To create a separate admission dimension table.
- To prepare a billing table.
- To create Patient ID and Admission ID.
- To connect the tables using relationships.
- To prepare the data for healthcare analysis and visualization.

DATA PREPARATION:

The healthcare dataset was imported into Power BI Desktop and opened in Power Query. The dataset contains information related to patients, medical conditions, hospital admissions, doctors, insurance providers, billing, medication and test results.

Power Query was used to transform the raw data before creating the final data model.



CLEANING CATEGORICAL DATA:

The categorical text columns were cleaned using Trim and Clean transformations.

These transformations help remove unnecessary spaces and unwanted characters from text values.

The cleaning was applied to columns such as:

- Name
- Gender
- Blood Type
- Medical Condition
- Doctor
- Hospital
- Insurance Provider
- Admission Type
- Medication
- Test Results

The screenshot displays a data transformation tool interface. The main table has 10 columns: Name, Age, Gender, Blood Type, Medical Condition, Date of Admission, and Doctor. The data is filtered to show rows where the Name column contains the text "Bobby JacksOn". The interface includes a toolbar with various data manipulation tools, a query editor, and a properties panel on the right.

	Name	Age	Gender	Blood Type	Medical Condition	Date of Admission	Doctor
1	Bobby JacksOn	30	Male	B-	Cancer	31-01-2024	Matthew Sml
2	Leslie TERly	62	Male	A+	Obesity	20-08-2019	Samantha Da
3	DafnINy sMITH	76	Female	A-	Obesity	22-09-2022	Tiffany Mitch
4	andriEw waTIS	28	Female	O+	Diabetes	18-11-2020	Kevin Wells
5	adriENNE beLI	43	Female	AB+	Cancer	19-09-2022	Kathleen Han
6	EMILY JOHNSOn	36	Male	A+	Asthma	20-12-2023	Taylor Newto
7	edwaRD EDWaRds	21	Female	AB-	Diabetes	03-11-2020	Kelly Olson
8	CHRISTINA MARTINEZ	20	Female	A+	Cancer	28-12-2021	Suzanne Thor
9	JASmiNe aGuilar	82	Male	AB+	Asthma	01-07-2020	Daniel Fergus
10	ChRISTopher BerG	58	Female	AB-	Cancer	23-06-2020	Heather Day
11	michElla danielS	72	Male	O+	Cancer	19-04-2020	John Duncan
12	aaRon MARTINEZ	38	Female	A-	Hypertension	13-08-2023	Douglas May
13	connOR HANSEN	75	Female	A+	Diabetes	12-12-2019	Kenneth Flet
14	rObERT bauer	68	Female	AB+	Asthma	22-05-2020	Theresa Free
15	brOCKe brady	44	Female	AB+	Cancer	08-10-2021	Roberta Stew
16	MS. nATalie gAMble	46	Female	AB-	Obesity	01-01-2023	Maria Dough
17	haley perkins	63	Female	A+	Arthritis	23-06-2020	Erica Spencer
18	mRS. jamIE cAMPBELl	38	Male	AB-	Obesity	08-03-2020	Justin Kim
19	LuKE BuRgEss	34	Female	A-	Hypertension	04-03-2021	Justin Moore
20	dANIEL schmidt	63	Male	B+	Asthma	15-11-2022	Denise Gall
21	TIMOTHY buRNs	67	Female	A-	Asthma	28-06-2023	Krista Smith
22	ChRISTOPHER BRIGHT	48	Male	B+	Asthma	21-01-2020	Gregory Smit
23	KathRYN Stewart	58	Female	O+	Arthritis	12-05-2022	Vanessa New
24	DR. EIEEN thomPsOn	59	Male	A+	Asthma	02-08-2021	Donna Marti
25							

15 COLUMNS. 999+ ROWS. Column profile based on top 1000 rows

PREVIEW DOWNLOADED AT 08:44 AM

The screenshot shows a data transformation tool interface. The main workspace displays a table with the following columns: Patient ID, Billing Amount, Room Number, Admission Type, Discharge Date, Medication, and Test Results. The table contains 25 rows of data. The interface includes a top menu bar with options like File, Home, Transform, Add Column, View, Tools, and Help. A left sidebar shows a query list with 'healthcare_dataset' selected. A right sidebar shows query settings and applied steps, including 'Cleaned Text'.

Patient ID	Billing Amount	Room Number	Admission Type	Discharge Date	Medication	Test Results
1	18856.28131	328	Urgent	02-02-2024	Paracetamol	Normal
2	33643.32729	265	Emergency	26-08-2019	Ibuprofen	Inconclusive
3	27955.09608	205	Emergency	07-10-2022	Aspirin	Normal
4	37909.78241	450	Elective	18-12-2020	Ibuprofen	Abnormal
5	14238.31781	458	Urgent	09-10-2022	Penicillin	Abnormal
6	48145.11095	389	Urgent	24-12-2023	Ibuprofen	Normal
7	19580.87234	389	Emergency	15-11-2020	Paracetamol	Inconclusive
8	45820.46272	277	Emergency	07-01-2022	Paracetamol	Inconclusive
9	50119.22279	316	Elective	14-07-2020	Aspirin	Abnormal
10	19784.63106	249	Elective	22-06-2021	Paracetamol	Inconclusive
11	12576.79561	394	Urgent	22-04-2020	Paracetamol	Normal
12	7999.58688	288	Urgent	05-09-2023	Lipitor	Inconclusive
13	43282.28336	134	Emergency	28-12-2019	Penicillin	Abnormal
14	33207.70663	309	Urgent	19-06-2020	Lipitor	Normal
15	40701.59923	182	Urgent	13-10-2021	Paracetamol	Normal
16	12263.35743	465	Elective	11-01-2023	Aspirin	Inconclusive
17	24499.8479	114	Elective	14-07-2020	Paracetamol	Normal
18	17440.46544	449	Urgent	02-04-2020	Paracetamol	Abnormal
19	18843.02302	260	Elective	14-03-2021	Aspirin	Abnormal
20	23762.20358	465	Elective	22-11-2022	Penicillin	Normal
21	42.51458855	115	Elective	02-07-2023	Aspirin	Normal
22	17695.91162	295	Urgent	09-02-2020	Lipitor	Normal
23	5998.102908	327	Urgent	10-06-2022	Lipitor	Inconclusive
24	25250.05243	119	Urgent	12-08-2021	Lipitor	Inconclusive
25	33333.33333	444	Emergency	05-05-2020	Paracetamol	Inconclusive

STANDARDIZING NAMES:

The Name column was standardized using the Capitalize Each Word transformation.

This makes the patient names follow a consistent capitalization format.

AGE GROUP:

A new column named Age Group was created using a conditional formula.

The age groups used are:

- 0–18
- 19–35
- 36–60
- 60+

Power Query Editor interface showing the 'Add Conditional Column' dialog box. The dialog is titled 'Add Conditional Column' and contains the following fields:

- New column name:** Age Group
- Column Name:** Age
- Operator:** is less than or equal to
- Value:** 18
- Output:** 0-18

The dialog also shows an 'Else If' clause with the following fields:

- Column Name:** Age
- Operator:** is less than or equal to
- Value:** 35
- Output:** 19-35

The dialog also shows an 'Else' clause with the following fields:

- Column Name:** Age
- Operator:** is less than or equal to
- Value:** 60
- Output:** 36-60

The background shows a table with columns: Date of Admission, Doctor, Hospital, Insurance Provider, Billing Amount, Room Number, and Admittance Status. The table contains data for 25 rows.

CREATING DIM_PATIENT:

The original healthcare dataset was duplicated in Power Query to create a separate patient dimension table. The new table was renamed Dim_Patient. From this table, only the patient-related columns such as Name, Age, Gender, Blood Type, and Age Group were retained. This created a separate table containing the required patient information.

Power Query Editor interface showing the 'Dim_Patient' table. The table has the following columns: Name, Age, Gender, Blood Type, Medical Condition, and Age Group. The table contains data for 25 rows.

The background shows a table with columns: Name, Age, Gender, Blood Type, Medical Condition, and Age Group. The table contains data for 25 rows.

CREATING PATIENT ID:

After creating the Dim_Patient table, an Index Column was added using the From 1 option. The newly created index column was renamed Patient ID. This provided a unique identifier for the patient records and was later used while merging the Billing table with Dim_Patient.

The screenshot shows the Power BI Desktop interface with the 'Dim_Patient' table loaded. The table has 25 rows and 6 columns: Name, Age, Gender, Blood Type, Age Group, and Patient ID. The 'Patient ID' column is highlighted, and the 'Renamed Columns' step is visible in the 'Applied Steps' pane on the right. The formula bar shows the query: `Table.RenameColumns(#"Added Index",{{"Index", "Patient ID"}})`.

Name	Age	Gender	Blood Type	Age Group	Patient ID
Bobby Jackson	30	Male	B-	19-35	1
Leslie Terry	62	Male	A+	60	2
Danny Smith	76	Female	A-	60	3
Andrew Watts	28	Female	O+	19-35	4
Adrienne Bell	43	Female	AB+	36-60	5
Emily Johnson	36	Male	A+	36-60	6
Edward Edwards	21	Female	AB-	19-35	7
Christina Martinez	20	Female	A+	19-35	8
Jasmine Aguilar	82	Male	AB+	60	9
Christopher Berg	58	Female	AB-	36-60	10
Michelle Daniels	72	Male	O+	60	11
Aaron Martinez	38	Female	A-	36-60	12
Connor Hansen	75	Female	A+	60	13
Robert Bauer	68	Female	AB+	60	14
Brooke Brady	44	Female	AB+	36-60	15
Ms. Natalie Gamble	46	Female	AB-	36-60	16
Haley Perkins	63	Female	A+	60	17
Mrs. Jamie Campbell	38	Male	AB-	36-60	18
Luke Burgess	34	Female	A-	19-35	19
Daniel Schmidt	63	Male	B+	60	20
Timothy Burns	67	Female	A-	60	21
Christopher Bright	48	Male	B+	36-60	22
Kathryn Stewart	58	Female	O+	36-60	23
Dr. Eileen Thompson	59	Male	A+	36-60	24
Paul Henderson	72	Female	AB+	60	25

CREATING DIM_ADMISSION & ADMISSION ID:

The original dataset was duplicated again to create the admission dimension table. This table was renamed Dim_Admission. The columns related to admission information were retained, including Admission Type, Doctor, Hospital, Medical Condition, Medication, Room Number, and Test Results. An Index Column was

then added using the From 1 option and renamed Admission ID.

	Hospital	Room Number	Admission Type	Medication	Test Results	Admission ID
1	Sons and Miller	328	Urgent	Paracetamol	Normal	0
2	Kim Inc	265	Emergency	Ibuprofen	Inconclusive	1
3	Cook PLC	205	Emergency	Aspirin	Normal	2
4	Hernandez Rogers and Vang,	450	Elective	Ibuprofen	Abnormal	3
5	White-White	458	Urgent	Penicillin	Abnormal	4
6	Nunez-Humphrey	389	Urgent	Ibuprofen	Normal	5
7	Group Middleton	389	Emergency	Paracetamol	Inconclusive	6
8	Powell Robinson and Valdez,	277	Emergency	Paracetamol	Inconclusive	7
9	Sons Rich and	316	Elective	Aspirin	Abnormal	8
10	Padilla-Walker	249	Elective	Paracetamol	Inconclusive	9
11	Schaefer-Porter	394	Urgent	Paracetamol	Normal	10
12	Lyons-Blair	288	Urgent	Lipitor	Inconclusive	11
13	Powers Miller, and Flores	134	Emergency	Penicillin	Abnormal	12
14	Rivera-Gutierrez	309	Urgent	Lipitor	Normal	13
15	Morris-Arellano	182	Urgent	Paracetamol	Normal	14
16	Cline-Williams	465	Elective	Aspirin	Inconclusive	15
17	Cervantes-Wells	114	Elective	Paracetamol	Normal	16
18	Torres, and Harrison Jones	449	Urgent	Paracetamol	Abnormal	17
19	Houston PLC	260	Elective	Aspirin	Abnormal	18
20	Hammond Ltd	465	Elective	Penicillin	Normal	19
21	Jones LLC	115	Elective	Aspirin	Normal	20
22	Williams-Davis	295	Urgent	Lipitor	Normal	21
23	Clark-Mayo	327	Urgent	Lipitor	Inconclusive	22
24	and Sons Smith	119	Urgent	Lipitor	Inconclusive	23
25	Wells-Perez	100	Emergency	Paracetamol	Inconclusive	24

CREATION OF BILLING TABLE:

A separate Billing table was created from the original healthcare dataset. The required patient, admission, and billing-related columns were retained in this table. The table included patient information along with admission details, Date of Admission, Insurance Provider, Billing Amount, and Discharge Date. This table was prepared so that the required patient and admission identifiers could be added through merging.

MERGING WITH DIM_PATIENT:

The Billing table was then merged with the Dim_Patient table to obtain the Patient ID. The matching patient information was selected from both tables, including Name, Age, Gender, Blood Type, and Age Group. A Left Outer join was selected for the merge. After completing the merge, the Dim_Patient column was expanded and only the Patient ID column was selected. This added the Patient ID to the Billing table.

Merge

Select a table and matching columns to create a merged table.

Billing

Name	1	Age	2	Gender	3	Blood Type	4	Medical Condition	Date of Admission	Doctor	
Bobby Jackson		30		Male		B-		Cancer	31-01-2024	Matthew Smith	Sons and
Leslie Terry		62		Male		A+		Obesity	20-08-2019	Samantha Davies	Kim Inc
Danny Smith		76		Female		A-		Obesity	22-09-2022	Tiffany Mitchell	Cook PLC
Andrew Watts		28		Female		O+		Diabetes	18-11-2020	Kevin Wells	Hernand

Dim_Patient

Name	1	Age	2	Gender	3	Blood Type	4	Age Group	5	Patient ID
Bobby Jackson		30		Male		B-		19-35		0
Leslie Terry		62		Male		A+		60+		1
Danny Smith		76		Female		A-		60+		2
Andrew Watts		28		Female		O+		19-35		3
Adrienne Bell		43		Female		AB+		36-60		4

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

► Fuzzy matching options

Estimating matches based on data previews

OK

Cancel

FileHomeTransformAdd ColumnViewToolsHelp

Close & ApplyNew SourceRecent SourcesEnter DataData source settingsData SourcesManage ParametersExport query resultsOutput DataRefresh PreviewAdvanced EditorQueryPropertiesChoose Remove ColumnsKeep Remove RowsReduce RowsSortSplit ColumnGroup ByData Type: TableUse First Row as HeadersReplace ValuesTransformMerge QueriesAppend QueriesCombine FilesCombine

CloseNew QueryData SourcesParametersOutput DataQueryPropertiesChoose Remove ColumnsKeep Remove RowsReduce RowsSortSplit ColumnGroup ByData Type: TableUse First Row as HeadersReplace ValuesTransformMerge QueriesAppend QueriesCombine FilesCombine

Queries [4]

healthcare_datasetDim_PatientDim_AdmissionBilling

Table.NestedJoin("#Renamed Columns", {"Name", "Age", "Gender", "Blood Type", "Age Group"}, Dim_Patient, {"Name", "Age", "Gender", "Blood

Admission Type

Discharge Date

Medication

Test Results

Age Group

Dim_Patient

100%Valid0%Error0%Empty

100%Valid0%Error0%Empty

100%Valid0%Error0%Empty

100%Valid0%Error0%Empty

ABC123

Search Columns to Expand

ExpandAggregate

(Select All Columns)

☐ Name☐ Age☐ Gender☐ Blood Type☐ Age Group☒ Patient ID

☐ Use original column name as prefix

OKCancel

1328Urgent02-02-2024ParacetamolNormal

2265Emergency26-08-2019IbuprofenInconclusive

3205Emergency07-10-2022AspirinNormal

4450Elective18-12-2020IbuprofenAbnormal

5458Urgent09-10-2022PenicillinAbnormal

6389Urgent24-12-2023IbuprofenNormal

7389Emergency15-11-2020ParacetamolInconclusive

8277Emergency07-01-2022ParacetamolInconclusive

9316Elective14-07-2020AspirinAbnormal

10249Elective22-06-2021ParacetamolInconclusive

11394Urgent22-04-2020ParacetamolNormal

12288Urgent05-09-2023LipitorInconclusive

13134Emergency28-12-2019PenicillinAbnormal

14309Urgent19-06-2020LipitorNormal

15182Urgent13-10-2021ParacetamolNormal

16465Elective11-01-2023AspirinInconclusive

17114Elective14-07-2020ParacetamolNormal

18449Urgent02-04-2020ParacetamolAbnormal

19260Elective14-03-2021AspirinAbnormal

20465Elective22-11-2022PenicillinNormal

21115Elective02-07-2023AspirinNormal

22295Urgent09-02-2020LipitorNormal

23327Urgent10-06-2022LipitorInconclusive

24119Urgent12-08-2021LipitorInconclusive

25126Emergency08-05-2020ParacetamolInconclusive

MERGING WITH DIM_ADMISSION:

After adding Patient ID, the Billing table was merged with the Dim_Admission table to obtain the Admission ID. The matching admission information, including Admission Type, Doctor, Hospital, Medical Condition, Medication, Room Number, and Test Results, was selected from both tables. A Left Outer join was used for the merge. The merged Dim_Admission column was then expanded and only Admission ID was selected. This added the Admission ID to the Billing table.

Merge

Select a table and matching columns to create a merged table.

Billing

	Room Number	4	Admission Type	5	Discharge Date	Medication	6	Test Results	7	Age Group	Patient ID
31	328		Urgent		02-02-2024	Paracetamol		Normal		19-35	0
29	265		Emergency		26-08-2019	Ibuprofen		Inconclusive		60+	1
08	205		Emergency		07-10-2022	Aspirin		Normal		60+	2
41	450		Elective		18-12-2020	Ibuprofen		Abnormal		19-35	3

Dim_Admission

2	Hospital	3	Room Number	4	Admission Type	5	Medication	6	Test Results	7	Admission ID
nith	Sons and Miller		328		Urgent		Paracetamol		Normal		0
avies	Kim Inc		265		Emergency		Ibuprofen		Inconclusive		1
hell	Cook PLC		205		Emergency		Aspirin		Normal		2
	Hernandez Rogers and Vang,		450		Elective		Ibuprofen		Abnormal		3

Join Kind

Left Outer (all from first, matching from second)

☐ Use fuzzy matching to perform the merge

▸ Fuzzy matching options

✓ The selection matches 56612 of 56612 rows from the first table.

OK

Cancel

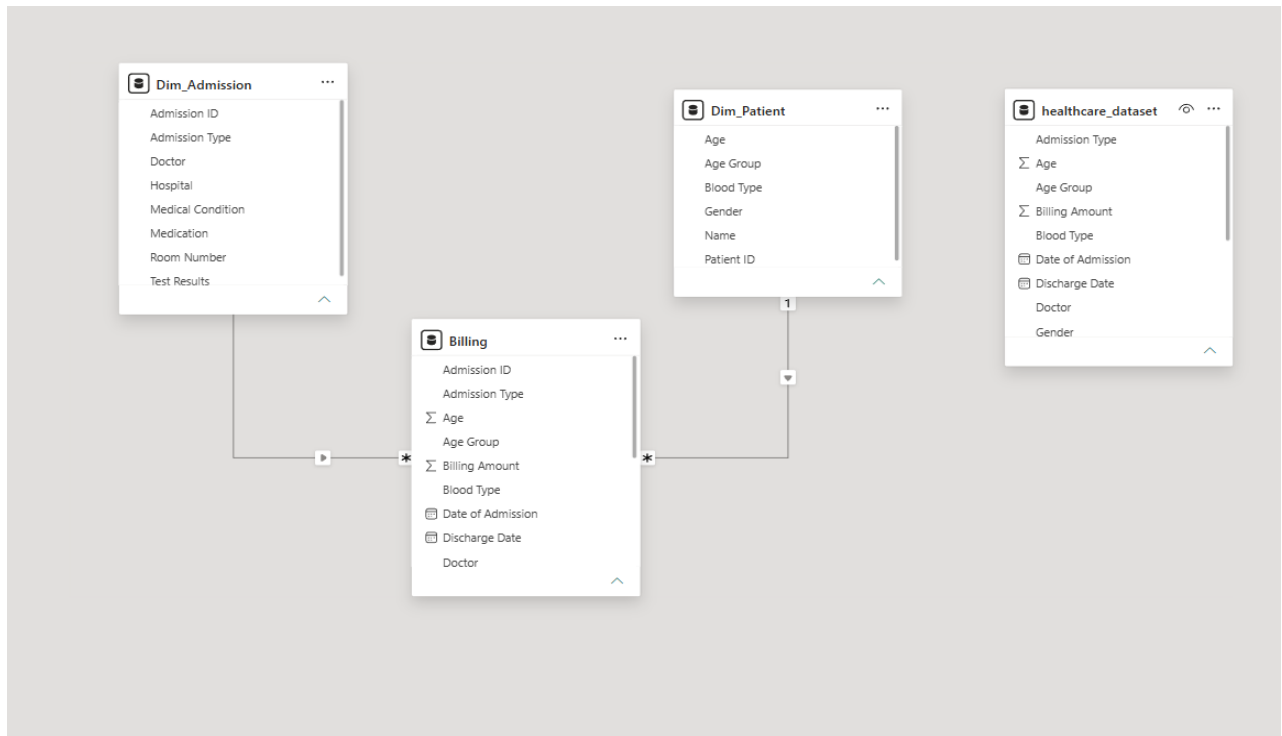
The screenshot shows the Power BI Desktop interface with a data table. The table has columns: Discharge Date, Medication, Test Results, Age Group, Patient ID, and Dim_Admission. A dialog box titled "Search Columns to Expand" is open, showing a list of columns to expand. The columns listed are: (Select All Columns), Medical Condition, Doctor, Hospital, Room Number, Admission Type, Medication, Test Results, and Admission ID. The "Admission ID" column is selected. The dialog box also has options for "Expand" (selected) and "Aggregate", and a checkbox for "Use original column name as prefix".

The screenshot shows the Power BI Desktop interface with the same data table. The 'Admission ID' column is now expanded, showing a list of admission IDs. The 'Query Settings' pane on the right is open, showing the 'Properties' and 'Applied Steps' for the 'Expanded Dim_Admission' query. The 'Properties' section shows the name 'Billing' and the source 'All Properties'. The 'Applied Steps' section shows a list of steps: Source, Promoted Headers, Changed Type, Trimmed Text, Cleaned Text, Capitalized Each Word, Added Conditional Column, Replaced Value, Added Custom, Removed Columns, Renamed Columns, Merged Queries, Expanded Dim_Patient, Merged Queries1, and Expanded Dim_Admission.

DATA MODEL:

After completing all the transformations, the final model contained three main tables: Dim_Patient, Dim_Admission, and Billing. The Billing table contained the required billing information along with Patient ID and Admission ID. The dimension tables provided the corresponding patient and admission information. The final model

was designed so that the dimension tables could be connected to the Billing table through their respective IDs.



CONCLUSION:

This project demonstrated how Power BI and Power Query can be used to clean, transform, and organize healthcare data. The original dataset was cleaned by applying text transformations and creating age groups. Separate patient and admission dimension tables were created, and the Billing table was prepared by merging the required information. The final data model provided an organized structure for performing healthcare data analysis in Power BI.